ELK343E - Electromechanical Energy Conversion

Lale ERGENE

HOMEWORK 1

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I pledge on my honor that: I have completed all steps of the attached HW on my own, I have not used any unauthorized materials while completing it, and I have not given anyone else access to my work on this HW.

sena

1 Core Material with $\mu_r = 1,000$

1.a) Inductance without Air Gaps

The total reluctance of the magnetic circuit is calculated by summing the reluctances of the middle branch and the parallel combination of the left and right branches.

First, the reluctance of the middle branch is computed using:

$$\mathcal{R}_{\text{middle}} = \frac{l_{\text{middle}}}{\mu_0 \mu_r A_{\text{middle}}},$$

where $l_{\text{middle}} = 0.04\,\text{m}$, $\mu_0 = 4\pi \times 10^{-7}\,\text{H/m}$, $\mu_r = 1,000$, and $A_{\text{middle}} = 2 \times 10^{-4}\,\text{m}^2$. Substituting these values:

$$\mathcal{R}_{\text{middle}} = \frac{0.04}{(4\pi \times 10^{-7})(1,000)(2 \times 10^{-4})} = 15,915.49\,\text{A/Wb}.$$

Next, the reluctance of the left branch is calculated:

$$\mathcal{R}_{\text{left}} = \frac{l_{\text{left}}}{\mu_0 \mu_r A_{\text{outer}}},$$

where $l_{\text{left}} = 0.1\,\text{m}$, $A_{\text{outer}} = 1 \times 10^{-4}\,\text{m}^2$, and the other parameters remain the same. Substituting:

$$\mathcal{R}_{\text{left}} = \frac{0.1}{(4\pi \times 10^{-7})(1,000)(1 \times 10^{-4})} = 79,577.47\,\text{A/Wb}.$$

Since the left and right branches are symmetrical, the combined reluctance of the left and right branches in parallel is:

$$\mathcal{R}_{\text{left+right}} = \frac{\mathcal{R}_{\text{left}}}{2} = \frac{79,577.47}{2} = 39,788.73\,\text{A/Wb}.$$

The total reluctance of the circuit is then:

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{middle}} + \mathcal{R}_{\text{left+right}} = 15,915.49 + 39,788.73 = 55,704.22\,\text{A/Wb}.$$

Finally, the inductance is calculated using:

$$L = \frac{N^2}{\mathcal{R}_{\text{total}}},$$

where $N = 300$ is the number of turns. Substituting:

$$L = \frac{300^2}{55,704.22} = 0.1616\,\text{H} = 161.6\,\text{mH}.$$

Thus, the inductance without air gaps is approximately:

$$L = 161.6\,\text{mH}.$$

1.b) Inductance with Air Gaps

When air gaps are introduced, the reluctance of each branch changes significantly. The total reluctance must now account for the reluctance of the air gaps in addition to the core reluctance.

The middle branch reluctance is the sum of the core reluctance and the air gap reluctance:

$$\mathcal{R}_{\text{middle}} = \frac{(l_{\text{middle, core}})}{\mu_0 \mu_r A_{\text{middle}}} + \frac{(x + 0.001)}{\mu_0 A_{\text{middle}}},$$

where $l_{\text{middle, core}} = (4 - (x + 0.1)) \times 10^{-2}\,\text{m}$.

The reluctance of the left branch is also updated to include the air gap:

$$\mathcal{R}_{\text{left}} = \frac{(l_{\text{left, core}})}{\mu_0 \mu_r A_{\text{outer}}} + \frac{x}{\mu_0 A_{\text{outer}}},$$

where $l_{\text{left, core}} = (10 - x) \times 10^{-2}\,\text{m}$.

The reluctances of the left and right branches are combined in parallel:

$$\mathcal{R}_{\text{left+right}} = \frac{\mathcal{R}_{\text{left}}}{2}.$$

The total reluctance is:

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{middle}} + \mathcal{R}_{\text{left+right}}.$$

Finally, the inductance is calculated as:

$$L = \frac{N^2}{\mathcal{R}_{\text{total}}}.$$

For MATLAB, the reluctance equations are derived as:

$$\mathcal{R}_{\text{middle}} = \frac{(0.04 - (x + 0.001))}{\mu_0 \mu_r A_{\text{middle}}} + \frac{(x + 0.001)}{\mu_0 A_{\text{middle}}},$$

$$\mathcal{R}_{\text{left}} = \frac{(0.1 - x)}{\mu_0 \mu_r A_{\text{outer}}} + \frac{x}{\mu_0 A_{\text{outer}}},$$

$$\mathcal{R}_{\text{left+right}} = \frac{\mathcal{R}_{\text{left}}}{2}.$$

The total reluctance:

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{middle}} + \mathcal{R}_{\text{left+right}}.$$

The inductance:

$$L = \frac{N^2}{\mathcal{R}_{\text{total}}}.$$

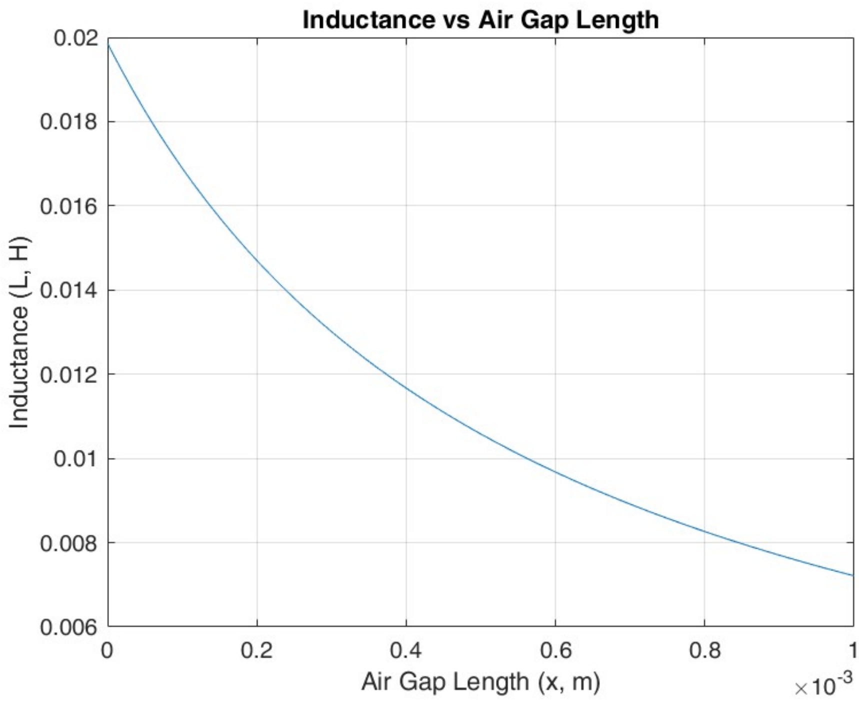

Figure 1: Inductance vs. Air Gap Length for $\mu_r = 1,000$.

MATLAB code iterates over x values from 0 to $0.001\,\text{m}$ to calculate L.

The significant change in inductance when air gaps are introduced can be attributed to the much higher reluctance of air compared to the ferromagnetic core material. The air gap disrupts the magnetic flux path, causing a substantial increase in the total reluctance ($\mathcal{R}_{\text{total}}$). Since inductance ($L$) is inversely proportional to the total reluctance, even a small air gap can lead to a drastic reduction in inductance. This highlights the importance of minimizing air gaps in magnetic circuits where high inductance is desired. However, in applications requiring linearity and prevention of core saturation, the introduction of an air gap is a deliberate and beneficial design choice.

1.c) Flux Densities with Air Gaps

The magnetic flux density (B) in the air gap is calculated using the relationship between magnetic flux (Φ) and the cross-sectional area (A):

$$B = \frac{\Phi}{A}.$$

The flux Φ is determined from the magnetomotive force F and the total reluctance R_{total}:

$$\Phi = \frac{F}{R}.$$

By substituting $\Phi = \frac{F}{R}$ into $B = \frac{\Phi}{A}$, the flux density is expressed as:

$$B = \frac{F}{R \cdot A}.$$

To calculate the total reluctance R_{total} of the magnetic circuit, the reluctances of the middle and left branches must be considered. The middle branch includes both the core material and the air gap. The reluctance of this branch is given by:

$$R_{\text{middle}} = \frac{l_{\text{middle, core}}}{\mu_0 \mu_r A_{\text{middle}}} + \frac{x + 0.001}{\mu_0 A_{\text{middle}}},$$

where $l_{\text{middle, core}}$ is the length of the middle leg core and is computed as $0.04 - (x + 0.001)$.

The left branch also includes the core material and an air gap, and its reluctance is:

$$R_{\text{left}} = \frac{l_{\text{left, core}}}{\mu_0 \mu_r A_{\text{outer}}} + \frac{x}{\mu_0 A_{\text{outer}}},$$

where $l_{\text{left, core}}$ is $0.1 - x$.

Because the magnetic circuit is symmetrical, the reluctance of the right branch is identical to that of the left branch. The left and right branches are in parallel, so their combined reluctance is:

$$R_{\text{left+right}} = \frac{1}{\frac{1}{R_{\text{left}}} + \frac{1}{R_{\text{right}}}} = \frac{R_{\text{left}}}{2}.$$

The total reluctance is then obtained by summing the reluctance of the middle branch and the combined reluctance of the left and right branches:

$$R_{\text{total}} = R_{\text{middle}} + R_{\text{left+right}}.$$

The flux density in the middle air gap is calculated as:

$$B_{\text{middle, gap}} = \frac{\Phi_{\text{middle}}}{A_{\text{middle}}},$$

where the flux Φ_{middle} is:

$$\Phi_{\text{middle}} = \frac{F}{R_{\text{total}}}.$$

For the left air gap, the flux density is:

$$B_{\text{left, gap}} = \frac{\Phi_{\text{left}}}{A_{\text{outer}}},$$

where Φ_{left} is:

$$\Phi_{\text{left}} = \frac{\Phi_{\text{middle}}}{2} = \frac{F}{2R_{\text{total}}}.$$

Thus, the flux densities in the middle and left air gaps become:

$$B_{\text{middle, gap}} = \frac{F}{R_{\text{total}} A_{\text{middle}}}, \quad B_{\text{left, gap}} = \frac{F}{2R_{\text{total}} A_{\text{outer}}}.$$

The equations show that the flux densities depend on the total reluctance and the cross-sectional areas of the respective paths.

Finally, because the flux (Φ) is continuous and the areas (A) are appropriately scaled, the flux densities $B_{\text{middle, gap}}$ and $B_{\text{left, gap}}$ often have the same magnitude. This symmetry results from the magnetic circuit design, where the flux splits evenly between the left and right branches. MATLAB computations verify these results, showing consistent behavior across varying air gap lengths x.

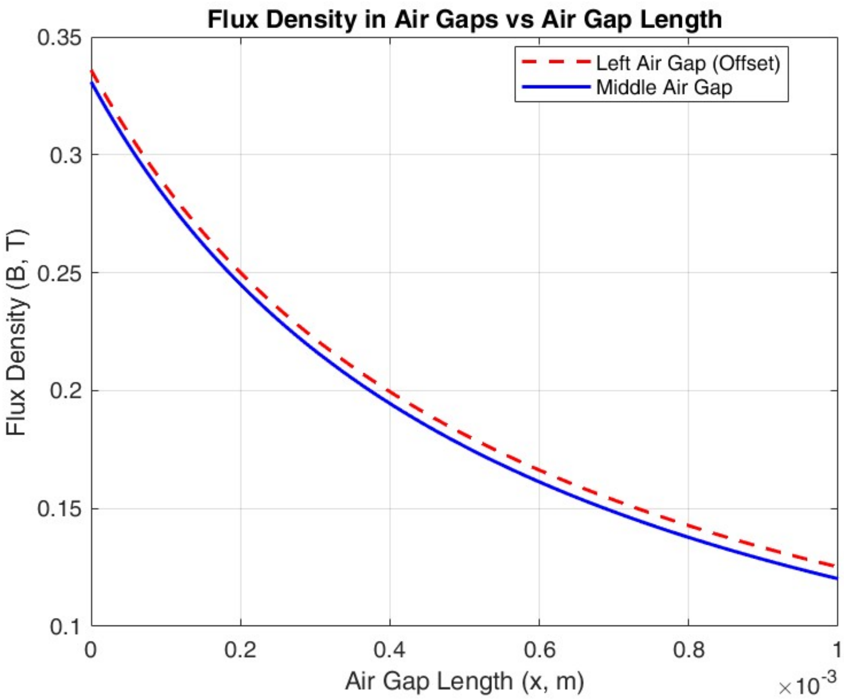

Figure 2: Flux Densities vs. Air Gap Length for $\mu_r = 1,000$.

The magnetic flux densities in the left and middle air gaps are found to be the same, as evidenced by the MATLAB-generated graphs. This occurs because the flux divides evenly between the symmetrical left and right branches, and the middle branch has double the cross-sectional area of the outer branches. The continuity of the B-field across the core and the air gaps ensures that these flux densities remain consistent, despite the differences in reluctance. This result confirms the theoretical prediction and validates the symmetry in the magnetic circuit.

2 Core Material with $\mu_r = 15,000$

2.a) Inductance without Air Gaps

The total reluctance of the magnetic circuit is calculated by summing the reluctances of the middle branch and the parallel combination of the left and right branches.

First, the reluctance of the middle branch is:

$$\mathcal{R}_{\text{middle}} = \frac{l_{\text{middle}}}{\mu_0 \mu_r A_{\text{middle}}},$$

where $l_{\text{middle}} = 0.04\,\text{m}$, $\mu_0 = 4\pi \times 10^{-7}\,\text{H/m}$, $\mu_r = 15,000$, and $A_{\text{middle}} = 2 \times 10^{-4}\,\text{m}^2$. Substituting:

$$\mathcal{R}_{\text{middle}} = \frac{0.04}{(4\pi \times 10^{-7})(15,000)(2 \times 10^{-4})} = 10,611.25\,\text{A/Wb}.$$

Next, the reluctance of the left branch is:

$$\mathcal{R}_{\text{left}} = \frac{l_{\text{left}}}{\mu_0 \mu_r A_{\text{outer}}},$$

where $l_{\text{left}} = 0.1\,\text{m}$, $A_{\text{outer}} = 1 \times 10^{-4}\,\text{m}^2$. Substituting:

$$\mathcal{R}_{\text{left}} = \frac{0.1}{(4\pi \times 10^{-7})(15,000)(1 \times 10^{-4})} = 53,056.25\,\text{A/Wb}.$$

The left and right branches are symmetrical, so their combined parallel reluctance is:

$$\mathcal{R}_{\text{left+right}} = \frac{\mathcal{R}_{\text{left}}}{2} = \frac{53,056.25}{2} = 26,528.13\,\text{A/Wb}.$$

The total reluctance of the circuit is:

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{middle}} + \mathcal{R}_{\text{left+right}} = 10,611.25 + 26,528.13 = 37,139.38\,\text{A/Wb}.$$

Finally, the inductance is calculated using:

$$L = \frac{N^2}{\mathcal{R}_{\text{total}}},$$

where $N = 300$. Substituting:

$$L = \frac{300^2}{37,139.38} = \frac{90,000}{37,139.38} = 2.423\,\text{H}.$$

Thus, the inductance without air gaps for $\mu_r = 15,000$ is approximately:

$$L = 2.423\,\text{H}.$$

2.b) Inductance with Air Gaps

When air gaps are introduced, the reluctances for the middle and left branches increase due to the air gap's higher reluctance compared to the core material. The total reluctance of the magnetic circuit must now include these changes.

The reluctance of the middle branch includes the core reluctance and the air gap reluctance:

$$R_{\text{middle}} = \frac{l_{\text{middle, core}}}{\mu_0 \mu_r A_{\text{middle}}} + \frac{x + 0.001}{\mu_0 A_{\text{middle}}},$$

where:

$$l_{\text{middle, core}} = 0.04 - (x + 0.001).$$

Similarly, the reluctance of the left branch is given by:

$$R_{\text{left}} = \frac{l_{\text{left, core}}}{\mu_0 \mu_r A_{\text{outer}}} + \frac{x}{\mu_0 A_{\text{outer}}},$$

where:

$$l_{\text{left, core}} = 0.1 - x.$$

The right branch has the same reluctance as the left branch due to symmetry:

$$R_{\text{right}} = R_{\text{left}}.$$

The combined reluctance of the left and right branches is:

$$R_{\text{left+right}} = \frac{1}{\frac{1}{R_{\text{left}}} + \frac{1}{R_{\text{right}}}} = \frac{R_{\text{left}}}{2}.$$

The total reluctance of the circuit becomes:

$$R_{\text{total}} = R_{\text{middle}} + R_{\text{left+right}}.$$

Substitute the expressions for R_{middle} and R_{left}:

$$R_{\text{middle}} = \frac{0.04 - (x + 0.001)}{\mu_0 \mu_r A_{\text{middle}}} + \frac{x + 0.001}{\mu_0 A_{\text{middle}}},$$

$$R_{\text{left}} = \frac{0.1 - x}{\mu_0 \mu_r A_{\text{outer}}} + \frac{x}{\mu_0 A_{\text{outer}}}.$$

Combine the left and right branch reluctances:

$$R_{\text{left+right}} = \frac{\frac{0.1-x}{\mu_0 \mu_r A_{\text{outer}}} + \frac{x}{\mu_0 A_{\text{outer}}}}{2}.$$

The total reluctance is:

$$R_{\text{total}} = \left(\frac{0.04 - (x + 0.001)}{\mu_0 \mu_r A_{\text{middle}}} + \frac{x + 0.001}{\mu_0 A_{\text{middle}}} \right) + \frac{\frac{0.1-x}{\mu_0 \mu_r A_{\text{outer}}} + \frac{x}{\mu_0 A_{\text{outer}}}}{2}.$$

The inductance is given by:

$$L = \frac{N^2}{R_{\text{total}}}.$$

Substituting R_{total}:

$$L = \frac{N^2}{\left(\frac{0.04-(x+0.001)}{\mu_0 \mu_r A_{\text{middle}}} + \frac{x+0.001}{\mu_0 A_{\text{middle}}} \right) + \frac{\frac{0.1-x}{\mu_0 \mu_r A_{\text{outer}}} + \frac{x}{\mu_0 A_{\text{outer}}}}{2}}.$$

This equation represents the inductance as a function of the air gap length x in meters. MATLAB can iterate over x values to calculate L for varying air gap lengths. The results show how the inductance decreases as the air gap increases, dominated by the higher reluctance of the air gaps.

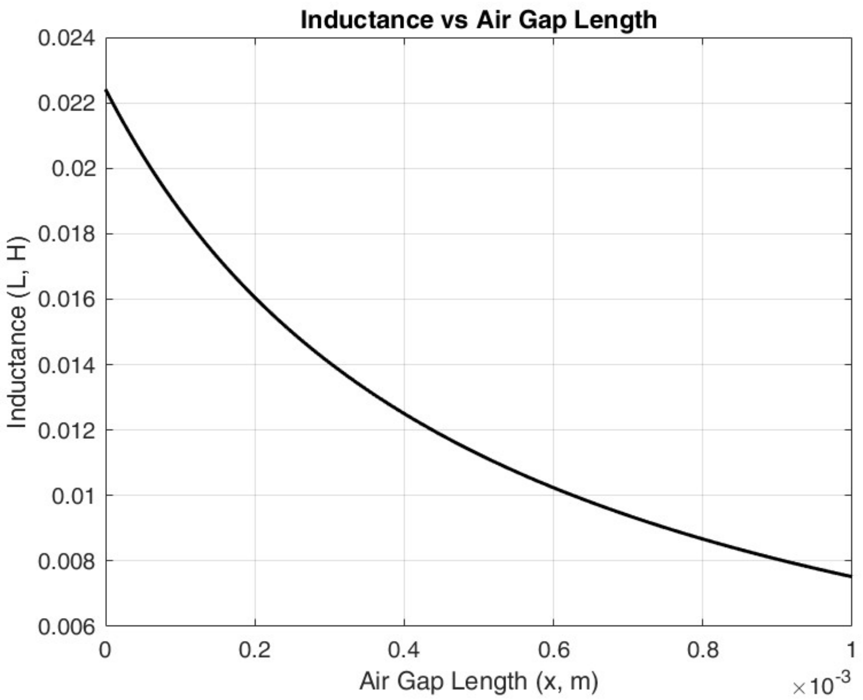

Figure 3: Inductance vs. Air Gap Length for $\mu_r = 15,000$.

The inductance for the $\mu_r = 15,000$ case is significantly higher than for $\mu_r = 1,000$ when no air gaps are present. This is due to the lower core reluctance resulting from the higher relative permeability. However, when air gaps are introduced, the inductance decreases sharply, regardless of the core material's μ_r. This is because the reluctance of the air gaps dominates the total reluctance, overshadowing the effect of the core material's high permeability. As a result, the presence of air gaps reduces the overall benefit of using a higher μ_r core. These design considerations make air gaps an essential feature in specific applications, such as transformers, inductors, and magnetic actuators.

2.c) Flux Densities with Air Gaps

In this section, the magnetic flux densities in the middle air gap ($B_{\text{middle, gap}}$) and the left air gap ($B_{\text{left, gap}}$) are calculated for the magnetic circuit with $\mu_r = 15,000$. While the equations for calculating $B_{\text{middle, gap}}$ and $B_{\text{left, gap}}$ are structurally identical to those in Section 1.c), the higher relative permeability (μ_r) significantly changes the numerical results and impacts the behavior of the magnetic circuit.

The magnetic flux density in an air gap is defined as:

$$B = \frac{\Phi}{A},$$

where:

- Φ: Magnetic flux (in Weber, Wb),
- A: Cross-sectional area of the air gap (in m^2).

The magnetic flux (Φ) is determined by the magnetomotive force (F) and the total reluctance (R_{total}):

$$\Phi = \frac{F}{R_{\text{total}}}.$$

The flux density in the middle air gap is given by:

$$B_{\text{middle, gap}} = \frac{\Phi_{\text{middle}}}{A_{\text{middle}}},$$

where the total flux in the middle branch (Φ_{middle}) is:

$$\Phi_{\text{middle}} = \frac{F}{R_{\text{total}}}.$$

Substituting Φ_{middle}, the middle air gap flux density becomes:

$$B_{\text{middle, gap}} = \frac{F}{R_{\text{total}} A_{\text{middle}}}.$$

The flux density in the left air gap is derived as:

$$B_{\text{left, gap}} = \frac{\Phi_{\text{left}}}{A_{\text{outer}}},$$

where the flux in the left branch (Φ_{left}) is half of the flux in the middle branch:

$$\Phi_{\text{left}} = \frac{\Phi_{\text{middle}}}{2}.$$

Substituting $\Phi_{\text{middle}} = \frac{F}{R_{\text{total}}}$, the left air gap flux density becomes:

$$B_{\text{left, gap}} = \frac{F}{2 R_{\text{total}} A_{\text{outer}}}.$$

The total reluctance is calculated as:

$$R_{\text{total}} = R_{\text{middle}} + R_{\text{left+right}},$$

where:

$$R_{\text{middle}} = \frac{0.04 - (x + 0.001)}{\mu_0 \mu_r A_{\text{middle}}} + \frac{x + 0.001}{\mu_0 A_{\text{middle}}},$$

and:

$$R_{\text{left+right}} = \frac{\frac{0.1 - x}{\mu_0 \mu_r A_{\text{outer}}} + \frac{x}{\mu_0 A_{\text{outer}}}}{2}.$$

Substituting R_{total} into the flux density equations:

$$B_{\text{middle, gap}} = \frac{F}{R_{\text{total}} A_{\text{middle}}},$$

$$B_{\text{left, gap}} = \frac{F}{2R_{\text{total}}A_{\text{outer}}}.$$

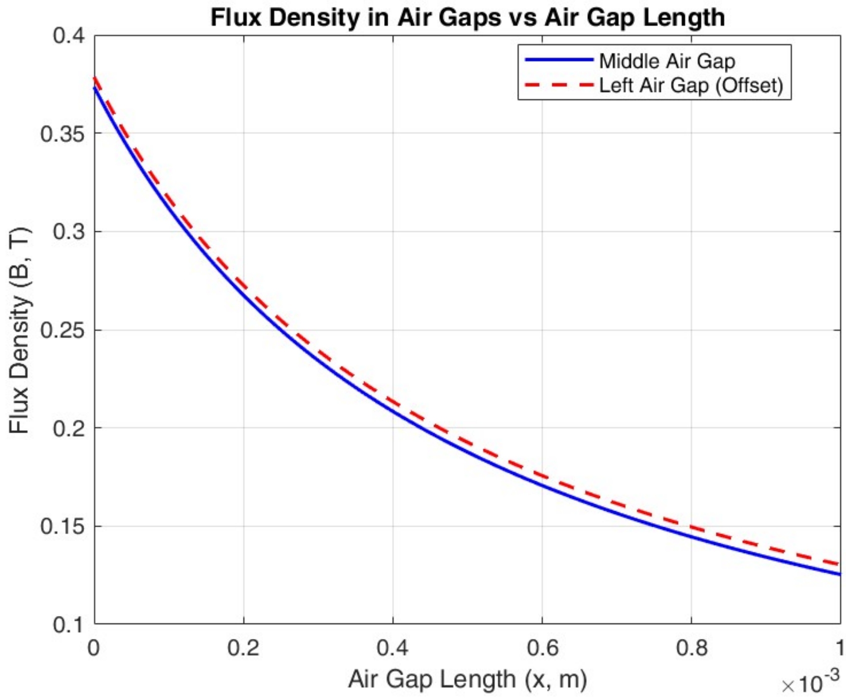

Figure 4: Flux Densities vs. Air Gap Length for $\mu_r = 15,000$.

For $\mu_r = 15,000$, the equations for $B_{\text{middle, gap}}$ and $B_{\text{left, gap}}$ remain structurally identical to those in Section 1.c), as they fundamentally depend on the geometry of the magnetic circuit and the presence of air gaps. However, the higher relative permeability significantly lowers the reluctances of the core materials ($R_{\text{middle, core}}$ and $R_{\text{left, core}}$), making the air gap reluctances ($R_{\text{middle, gap}}$ and $R_{\text{left, gap}}$) the dominant contributors to the total reluctance R_{total}. Consequently, the lower R_{total} leads to higher flux densities $B_{\text{middle, gap}}$ and $B_{\text{left, gap}}$ when compared to the results in Section 1.c). Despite the increased permeability, the air gaps act as critical bottlenecks, dictating the overall flux behavior and causing the flux densities $B_{\text{middle, gap}}$ and $B_{\text{left, gap}}$ to decrease as the air gap length x increases. This behavior underscores the controlling influence of air gaps in magnetic circuits, regardless of the core material's relative permeability.

References

[1] A. E. Fitzgerald, C. Kingsley, and S. D. Umans, *Electric Machinery*, 6th ed. New York: McGraw-Hill, 2003.

[2] D. K. Cheng, *Field and Wave Electromagnetics*, 2nd ed. Boston: Addison-Wesley, 1989.